$$\begin{cases} \partial_t f & = \sigma_x \Delta_x f - \mathrm{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) & = f_0 \end{cases}$$

- What happens as $t \to \infty$?

- Part 2 (Rakotomalala & dW, in review, 2026):

- **Theorem.** There is a universal $c > 0$ such that if $\sigma_x < c\mathrm{Pe}/2\pi$ and σ_θ small enough, then there is a value of χ at which there exist two branches representing two families of stationary solutions: one constant in one spatial x-variable (lanes) and one swap-symmetric: $f(x_1, x_2, \theta) = f(-x_2, -x_2, -\theta - \pi/2)$ (spots)

- Existence of critical $\tau = \tau^*$ with super- to subcritical transitions

- For lanes and spots separately and $0 < \tau_s^* < \tau_l^*$

- $\tau < \tau_s^*$ spots are locally nonlinearly stable (up to translation)

Ant dynamics

Ant dynamics

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Ant dynamics

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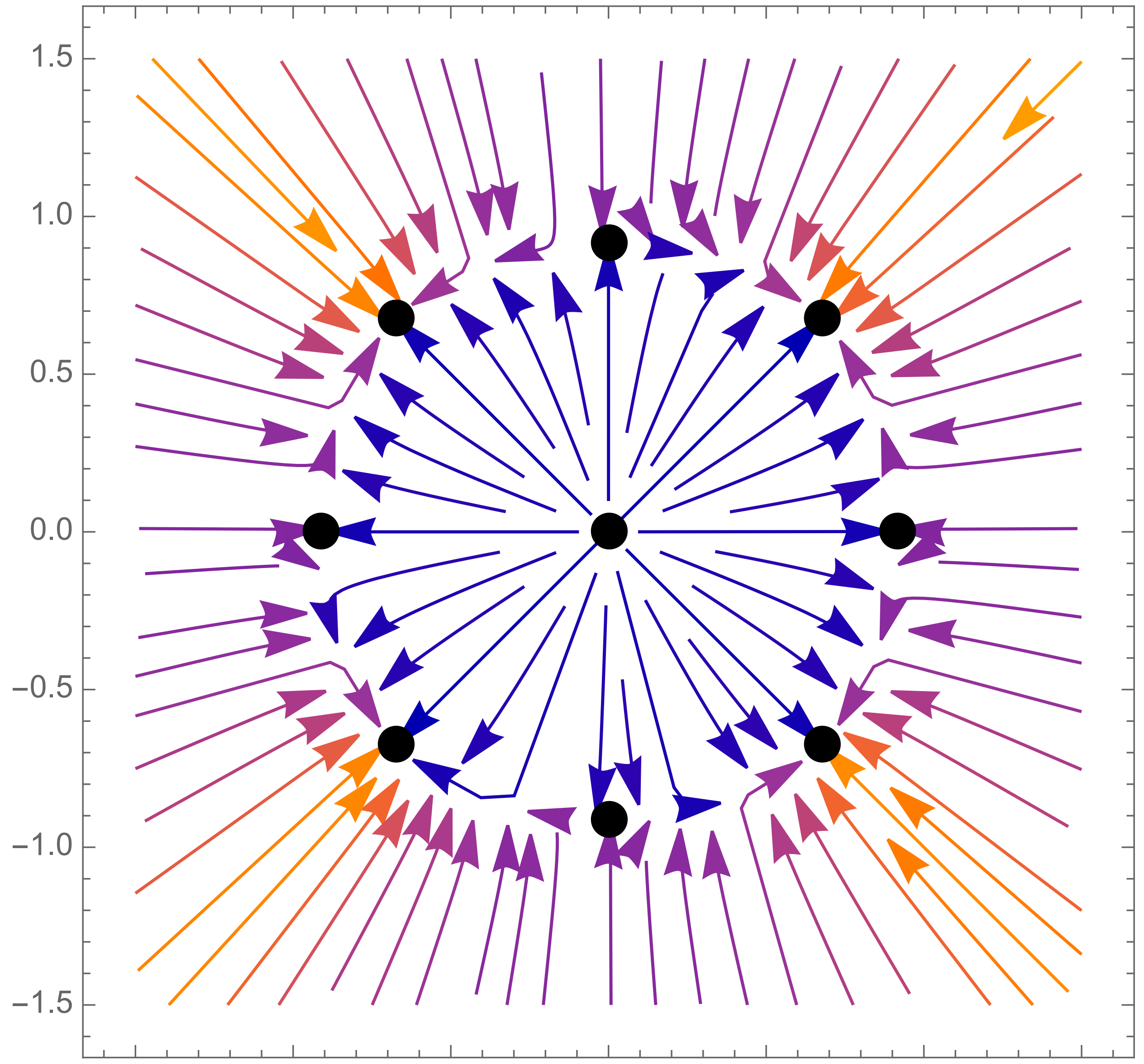
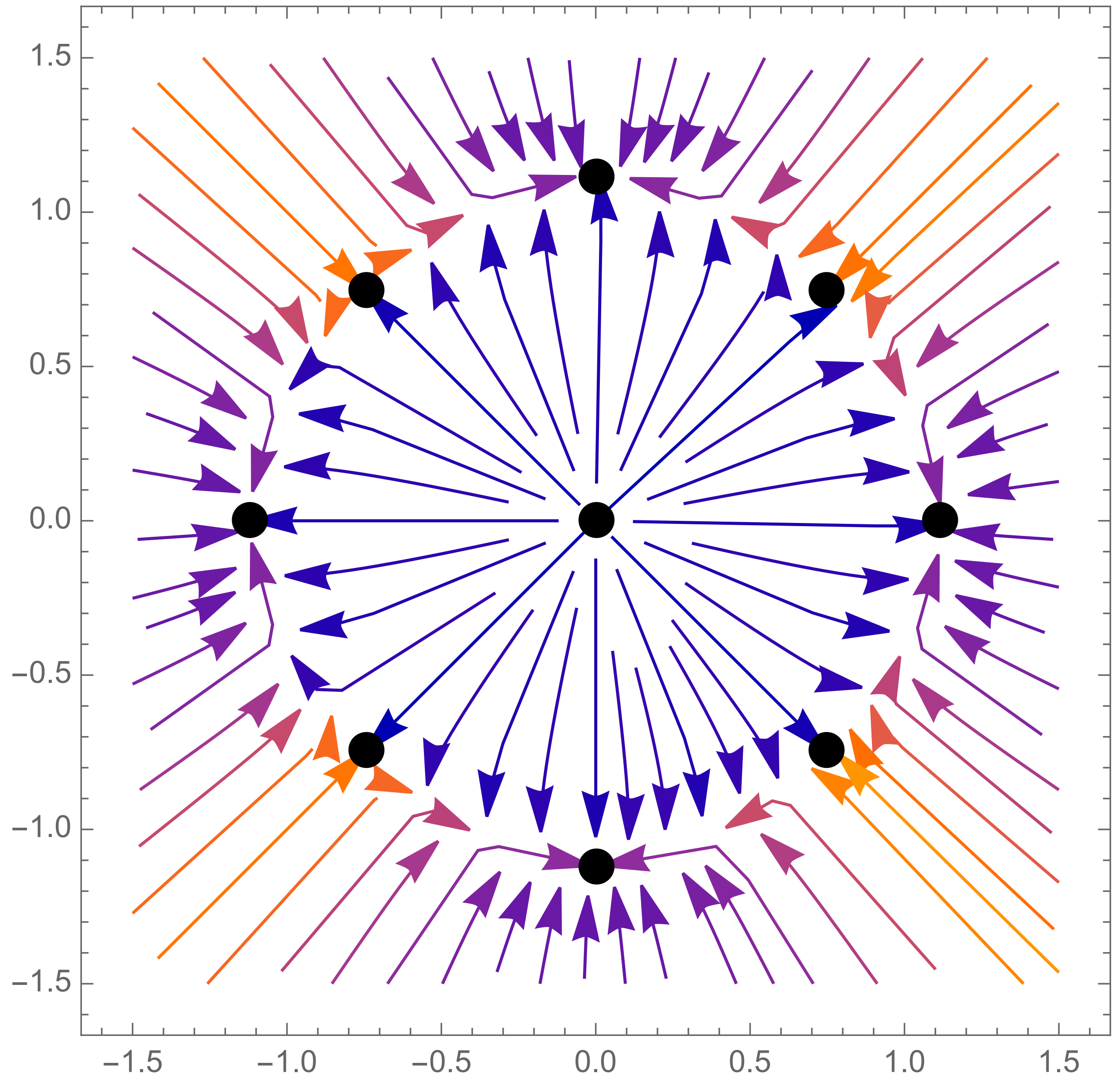
- What happens as $t \to \infty$?
- Part 2 (Rakotomalala & dW, in review, 2026):
  - Existence of two families of non-trivial stationary states
  - Existence of critical $\tau = \tau^*$ super- to subcritical



$\tau < \tau^*$



$\tau > \tau^*$